CarDekho Used Car Price Analysis

Depreciation, Value Retention & Regression Modeling

Dataset: 15K+ used-car listings

Tools: Python, Plotly, Scikit-learn, Power BI, DAX

Best overall model: Gradient Boosting

Best R^2: 0.8086

1. Business Problem

A used-car marketplace needs a consistent way to explain resale value, understand depreciation, and estimate reasonable listing prices from vehicle characteristics. This project analyzes CarDekho listing data to identify major price drivers, compare depreciation patterns, and evaluate regression models for price estimation.

2. Data Preparation

- Removed the unnecessary index column and 2 duplicate rows.
- Corrected two invalid zero-seat records: Honda City and Nissan Kicks were set to 5 seats.
- Removed extreme km_driven observations above 300,000 km for modeling stability.
- Created derived features including selling price in lakh, kilometres per year, age groups, km bands, and power per litre.

3. Key Analytical Findings

Relationship	Correlation	Interpretation
Max Power vs Price	0.751	Strong positive relationship
Engine vs Price	0.587	Strong positive relationship
Mileage vs Price	-0.306	Negative association
Vehicle Age vs Price	-0.241	Age matters more than km alone
KM Driven vs Price	-0.121	Weak negative association
KM Driven vs Vehicle Age	0.490	Moderate positive relationship

4. Transmission Pricing

Automatic vehicles had a median listing price of ₹10.5 lakh versus ₹5.0 lakh for manual vehicles, a raw premium of 110%. This is an observational market difference, not the causal value of automatic transmission, because automatic vehicles may also be newer, higher-powered, or more concentrated in premium brands.

5. Depreciation & Value Retention

Depreciation is interpreted as cross-sectional market depreciation because the dataset contains vehicles at different ages rather than repeated observations of the same vehicle from purchase to resale. Value retention was normalized using the median used-market price of one-year-old vehicles as the baseline.

- Median used-car prices generally declined as vehicle age increased.
- In the initial 5-year comparison with sufficient observations, Maruti retained about 84.8% and Hyundai about 82.3% of their one-year-old median used-market value.
- The analysis should not be interpreted as exact historical depreciation from original MSRP.

6. Regression Model Comparison

Model	R^2	MAE	RMSE
Linear Regression	0.5277	₹2.19L	₹7.53L
Log Linear Regression	0.7083	₹1.57L	₹5.92L
Random Forest	0.7670	₹1.16L	₹5.29L
Gradient Boosting	0.8086	₹1.29L	₹4.80L

Gradient Boosting was selected as the strongest overall model because it achieved the highest R^2 and lowest RMSE. Random Forest achieved the lowest MAE.

7. Power BI Dashboard

Dashboard Page	Purpose
Page 1 - Market Overview	Market KPIs, top brands, median pricing, fuel mix, transmission mix, price ranges
Page 2 - Depreciation & Value Retention	Price by vehicle age, age-group pricing, five-year retention, market depreciation
Page 3 - Regression & Pricing Intelligence	Model KPIs, performance comparison, feature importance, actual vs predicted by brand, error by brand

8. Business Recommendations

- Use age, max power, engine capacity, brand, and transmission as core pricing inputs.
- Interpret kilometres driven together with vehicle age rather than independently.
- Prefer nonlinear regression models for marketplace price estimation.
- Track prediction error by brand and consider specialized models where error remains high.
- Use depreciation and value-retention analysis to support seller pricing guidance and buyer transparency.

9. Limitations

- Listing price may differ from final transaction price.
- Original purchase price/MSRP is unavailable.
- Depreciation is cross-sectional, not longitudinal.
- Vehicle condition, accident history, trim, service history, and location are not fully represented.